no wonder that speculations on extra dimensions took a distant backseat during these productive and heady times. With physicists exploring powerful quantum methods, the implications of which gave rise to experimentally testable predictions, there was little interest in the mere possibility that the universe might be a vastly different place on length scales far too small to be probed by even the most powerful of instruments.

But sooner or later, bull markets lose steam. By the late 1960s and early 1970s the theoretical structure of the standard model was in place. By the late 1970s and early 1980s many of its predictions had been verified experimentally, and most particle physicists concluded that it was just a matter of time before the rest were confirmed as well. Although a few important details remained unresolved, many felt that the major questions concerning the strong, weak, and electromagnetic forces had been answered.

The time was finally ripe to return to the grandest question of all: the enigmatic conflict between general relativity and quantum mechanics. The success in formulating a quantum theory of three of nature's forces emboldened physicists to try to bring the fourth, gravity, into the fold. Having pursued numerous ideas that all ultimately failed, the mind-set of the community became more open to comparatively radical approaches. After being left for dead in the late 1920s, Kaluza-Klein theory was resuscitated.

Modern Kaluza-Klein Theory

The understanding of physics had significantly changed and substantially deepened in the six decades since Kaluza's original proposal. Quantum mechanics had been fully formulated and experimentally verified. The strong and the weak forces, unknown in the 1920s, had been discovered and were largely understood. Some physicists suggested that Kaluza's original proposal had failed because he was unaware of these other forces and had therefore been too conservative in his revamping of space. More forces meant the need for even more dimensions. It was argued that a single new, circular dimension, although able to show hints of a connection between general relativity and electromagnetism, was just not enough.

By the mid-1970s, an intense research effort was underway, focusing on higher-dimensional theories with numerous curled-up